

BioLitRecs: Retrieval and Similarity Exploration for Cancer Biomarker Literature

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Abstract

Cancer biomarker research generates an extensive and evolving body of literature in which individual proteins are studied across therapeutic, prognostic, and mechanistic contexts. Conclusions regarding biomarker efficacy or clinical relevance may shift over time and often include supportive, negative, or inconclusive findings. Traditional keyword search retrieves documents but does not organize evidence by stance, temporal evolution, or structural context. This project proposes BioLitRecs, a retrieval and recommendation system built over a curated subset of cancer biomarker abstracts. The system integrates classical information retrieval methods, distributional word embeddings, stance detection, entity recognition, temporal modeling, and structural metadata derived from AlphaFold. The goal is to construct an interpretable and structured framework for exploring cancer biomarker literature.

1 Introduction

Cancer biomarkers are typically identified through genomic sequencing, mutation frequency analysis, transcriptomic profiling, or proteomic assays. Proteins such as EGFR [10], HER2 [8], KRAS [1], BRCA1 [4], TP53 [3], and PD-1 [9] play central roles in tumor progression, immune regulation, DNA repair, and targeted therapy development. As research accumulates, findings concerning a biomarker’s prognostic value or therapeutic response may shift, refine, or contradict earlier reports.

PubMed provides broad access to biomedical literature but primarily supports keyword-based retrieval. It does not structure results by reported stance, highlight conflicting evidence, or provide mechanisms for exploring how conclusions evolve over time. Recent advances in protein structure prediction, particularly AlphaFold [2], have further enriched the context in which biomarker proteins are interpreted. Although AlphaFold does not discover biomarkers directly, structural availability may influence downstream mechanistic research.

BioLitRecs aims to integrate retrieval, stance modeling, entity recognition, temporal analysis, and structural metadata into a unified exploratory system focused on cancer biomarker literature.

2 Dataset Construction

We will select approximately fifty cancer biomarkers from established biomedical resources and review articles. These biomarkers will represent diverse functional categories including growth signaling, tumor suppression, immune checkpoint regulation, and DNA repair.

For each biomarker, structured PubMed queries will combine the biomarker name with cancer-related terms such as therapy, mutation, inhibition, survival, prognosis, or resistance. For each

biomarker, we will sample roughly twenty to thirty abstracts while ensuring representation across:

- supportive findings (evidence suggesting therapeutic benefit or positive association),
- negative findings (evidence reporting lack of effect or adverse association),
- inconclusive or mixed findings.

This produces a dataset of approximately one thousand to fifteen hundred abstracts. For each paper, we will store abstract text, publication year, and PubMed identifier. Considerable time will be devoted to ensuring dataset balance and representativeness, as this directly influences retrieval behavior and recommendation quality.

3 Retrieval and Vectorization

All abstracts will undergo preprocessing including tokenization, normalization, and stopword removal. Documents will be represented using one of three vector space approaches:

- TF-IDF weighting [7],
- BM25 probabilistic ranking [6],
- Word2Vec embeddings [5].

Vectorization and embedding training will require parameter tuning and empirical validation to determine which representation produces the most coherent ranking behavior.

4 Stance Modeling

Each abstract will be assigned a coarse stance label: supportive, negative, or inconclusive. Labeling will be performed using a curated lexicon of indicative phrases such as “improves survival,” “no significant association,” or “associated with toxicity.” A subset of papers will be manually reviewed to validate labeling consistency.

Stance information allows the system to:

- group studies reporting similar conclusions,
- highlight conflicting evidence,
- analyze how stance distribution changes over time.

5 Entity and Structural Context

Biomarker names and synonyms will be compiled from curated protein databases such as UniProt. Drug associations mentioned in abstracts will also be recorded.

For each biomarker protein, we will record whether an AlphaFold predicted structure is available [2]. Structural availability will serve as contextual metadata and may support exploratory analysis of whether structurally characterized proteins exhibit different evidence patterns.

6 Similarity Framework

Similarity between two objects, either papers or biomarkers, is computed as a weighted combination of interpretable components:

$$\text{sim}(x, y) = \alpha \text{sim}_{\text{text}}(x, y) + \beta \text{sim}_{\text{stance}}(x, y) + \gamma \text{sim}_{\text{entity}}(x, y) + \delta \text{sim}_{\text{time}}(x, y) + \epsilon \text{sim}_{\text{structure}}(x, y).$$

Text similarity measures topical overlap using vector representations. Stance similarity reflects agreement in reported conclusions. Entity similarity captures overlap in referenced biomarkers or drugs. Temporal similarity models proximity in publication year, enabling analysis of evidence evolution rather than assuming newer studies are inherently correct. Structural similarity records shared structural metadata availability.

Users may adjust weights to explore literature under different similarity assumptions.

7 User Interface

The interface will allow users to:

- select biomarkers of interest,
- filter by publication year range,
- retrieve ranked results,
- adjust similarity weights dynamically,
- issue new search queries.

The design emphasizes exploratory analysis and transparency in similarity weighting.

8 Evaluation

Evaluation will include:

- Precision and recall for selected benchmark queries.
- Manual validation of stance labels.
- Qualitative assessment of recommendation coherence.
- Analysis of stance distribution over time for selected biomarkers.

9 Timeline and Teamwork

All team members will collaborate across phases.

Phase One: Biomarker Selection and Paper Discovery. Significant effort will be devoted to structured PubMed querying, dataset balancing, and metadata validation.

Phase Two: Vectorization and Retrieval Infrastructure. Implementation of TF-IDF, BM25, and Word2Vec representations, including parameter tuning and evaluation.

Phase Three: Stance and Entity Integration. Development and validation of stance labeling rules and entity detection.

Phase Four: Recommendation and Interface Integration. Integration of similarity components, calibration of weights, and iterative testing of recommender behavior.

Phase Five: Evaluation and Documentation. Experimental validation and preparation of final deliverables.

10 Conclusion

BioLitRecs combines structured dataset construction, classical information retrieval, embedding based semantic modeling, stance detection, temporal analysis, and structural context into an interpretable framework for exploring cancer biomarker literature. By emphasizing adjustable similarity modeling and evidence evolution, the system supports structured navigation of complex biomedical findings.

References

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